Class 19: Mini Project Cancer Mutation Analysis

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Background 1

Background

To identify somatic mutations in a tumor, DNA from the tumor is sequenced and compared to DNA from normal tissue in the same individual using variant calling algorithms.

Comparison of tumor sequences to those from normal tissue (rather than 'the human genome') is important to ensure that the detected differences are not germline mutations.

To identify which of the somatic mutations leads to the production of aberrant proteins, the location of the mutation in the genome is inspected to identify non-synonymous mutations (i.e. those that fall into protein coding regions and change the encoded amino acid).

I downloaded my sequence from the class website and moved it to my working directory:

> Q1. What protein do these sequences correspond to? (Give both full gene/protein name and official symbol).

From a **BLASTp** search against human refseq I see that it is:

- Official Symbol: JAK2
- Official Full Name: Janus kinase 2

```
library(bio3d)

seq <- read.fasta("A18541014_mutant_seq.fa")
seq
```

```
                1         .         .         .         .         .        60
wt_healthy      MGMACLTMTEMEGTSTSSIYQNGDISGNANSMKQIDPVLQVYLYHSLGKSEADYLTFPSG
mutant_tumor    MGMACLTMTEMEGTSTSSIYQNGDISGNANSMKQIDPVLQVYLYHSLGKSEADYLTFPSG
                ************************************************************
                1         .         .         .         .         .        60

               61         .         .         .         .         .       120
wt_healthy      EYVAEEICIAASKACGITPVYHNMFALMSETERIWYPPNHVFHIDESTRHNVLYRIRFYF
mutant_tumor    EYVAEEICIAASKACGITPVYHNMFALMSETERIWYPPNHVFHIDESTRHNVLYRIRFYF
                ************************************************************
               61         .         .         .         .         .       120

              121         .         .         .         .         .       180
wt_healthy      PRWYCSGSNRAYRHGISRGAEAPLLDDFVMSYLFAQWRHDFVHGWIKVPVTHETQEECLG
mutant_tumor    PRWYCSGSNRAYRHGISRGAEAPLLDDFVMSYLFAQWRHDFVHGWIKVPVTHETQEECLG
                ************************************************************
              121         .         .         .         .         .       180

              181         .         .         .         .         .       240
wt_healthy      MAVLDMMRIAKENDQTPLAIYNSISYKTFLPKCIRAKIQDYHILTRKRIRYRFRRFIQQF
mutant_tumor    MAVLDMMRIAKENDQTPLAIYNSISYKTFLPKCIRAKIQDYHILTRKRIRYRFRRFIQQF
                ************************************************************
              181         .         .         .         .         .       240

              241         .         .         .         .         .       300
wt_healthy      SQCKATARNLKLKYLINLETLQSAFYTEKFEVKEPGSGPSGEEIFATIIITGNGGIQWSR
mutant_tumor    SQCKATARNLKLKYLINLETLQSAFYTEKFEVKEPGSGPSGEEIFATIIITGNGGIQWSR
                ************************************************************
              241         .         .         .         .         .       300

              301         .         .         .         .         .       360
wt_healthy      GKHKESETLTEQDLQLYCDFPNIIDVSIKQANQEGSNESRVVTIHKQDGKNLEIELSSLR
mutant_tumor    GKHKESETLTEQDLQLYCDFPNIIDVSIKQANQEGSNESRVVTIHKQDGKNLEIELSSLR
                ************************************************************
              301         .         .         .         .         .       360

              361         .         .         .         .         .       420
wt_healthy      EALSFVSLIDGYYRLTADAHHYLCKEVAPPAVLENIQSNCHGPISMDFAISKLKKAGNQT
mutant_tumor    EALSFVSLIDGYYRLTADAHHYLCKEVAPPAVLENIQSNCHGPISMDFAISKLKKAGNQT
                ************************************************************
              361         .         .         .         .         .       420

              421         .         .         .         .         .       480
```

```
wt_healthy     GLYVLRCSPKDFNKYFLTFAVERENVIEYKHCLITKNENEEYNLSGTKKNFSSLKDLLNC
mutant_tumor   GLYVLRCSPKDFNKYFLTFAVERENVIEYKHCLITKNENEEYNLSGTKKNFSSLKDLLNC
               ************************************************************
             421        .         .         .         .         .        480

             481        .         .         .         .         .        540
wt_healthy     YQMETVRSDNIIFQFTKCCPPKPKDKSNLLVFRTNGVSDVPTSPTLQRPTHMNQMVFHKI
mutant_tumor   YQMETVRSDNIIFQFTKCCPPKPKDKSNLLVFRTNGVSDVPTSPTLQRPTHMNQMVFHKI
               ************************************************************
             481        .         .         .         .         .        540

             541        .         .         .         .         .        600
wt_healthy     RNEDLIFNESLGQGTFTKIFKGVRREVGDYGQLHETEVLLKVLDKAHRNYSESFFEAASM
mutant_tumor   RNEDLIFNESLGQGTFTKIFKGVRVEVGDYGQLHETEVLLKVLDKAHRNYSESFFEAASM
               ************************ ***********************************
             541        .         .         .         .         .        600

             601        .         .         .         .         .        660
wt_healthy     MSKLSHKHLVLNYGVCVCGDENILVQEFVKFGSLDTYLKKNKNCINILWKLEVAKQLAWA
mutant_tumor   MSKLSHKHLVLNYGVCVCGDENILVQEFVKFGELDTYLKKNKNCINILWKLEVAKQLAWA
               ******************************** **************************
             601        .         .         .         .         .        660

             661        .         .         .         .         .        720
wt_healthy     MHFLEENTLIHGNVCAKNILLIREEDRKTGNPPFIKLSDPGISITVLPKDILQERIPWVP
mutant_tumor   MHFLEENTLIHGNVCAKNILLIREEDRKTGNPPFIKLSDPGISITVLRKDILQERIPWVP
               ********************************************** ************
             661        .         .         .         .         .        720

             721        .         .         .         .         .        780
wt_healthy     PECIENPKNLNLATDKWSFGTTLWEICSGGDKPLSALDSQRKLQFYEDRHQLPAPKWAEL
mutant_tumor   PECIENPKNLNLATDKWSFGTTLWEICSGGDKPLSALDSQRKLQFYEDRHQLPAPKWAEL
               ************************************************************
             721        .         .         .         .         .        780

             781        .         .         .         .         .        840
wt_healthy     ANLINNCMDYEPDFRPSFRAIIRDLNSLFTPDYELLTENDMLPNMRIGALGFSGAFEDRD
mutant_tumor   ANLINNCMDYEPDFRPSFRAIIRDLNSLFTPDYELLTENDMLPNMRIGALGFSGAFEDRD
               ************************************************************
             781        .         .         .         .         .        840

             841        .         .         .         .         .        900
wt_healthy     PTQFEERHLKFLQQLGKGNFGSVEMCRYDPLQDNTGEVVAVKKLQHSTEEHLRDFEREIE
```

```
mutant_tumor   PTQFEERHLKFLQQLGKGNFGSVEMCRYDPLQDNTGEVVAVKKLQHSTEEHLRDFEREIE
               ************************************************************
             841        .         .         .         .         .         900

             901        .         .         .         .         .         960
wt_healthy     ILKSLQHDNIVKYKGVCYSAGRRNLKLIMEYLPYGSLRDYLQKHKERIDHIKLLQYTSQI
mutant_tumor   ILKSLQHDNIVKYKGVCYSAGRRNLKLIMEYLPYGSLRDYLQKHKERIDHIKLLQYTSQI
               ************************************************************
             901        .         .         .         .         .         960

             961        .         .         .         .         .         1020
wt_healthy     CKGMEYLGTKRYIHRDLATRNILVENENRVKIGDFGLTKVLPQDKEYYKVKEPGESPIFW
mutant_tumor   CKGMEYLGTKRYIHRDLATRNILVENENRVKIGDFGLTKVLPQDKEYYKVKEPGESPIFW
               ************************************************************
             961        .         .         .         .         .         1020

            1021        .         .         .         .         .         1080
wt_healthy     YAPESLTESKFSVASDVWSFGVVLYELFTYIEKSKSPPAEFMRMIGNDKQGQMIVFHLIE
mutant_tumor   YAPESLTESKFSVASDVWSFGVVLYELFTYIEKSKSPPAEFMRMIGNDKQGQMIVFHLIE
               ************************************************************
            1021        .         .         .         .         .         1080

            1081        .         .         .         .         . 1132
wt_healthy     LLKNNGRLPRPDGCPDEIYMIMTECWNNNVNQRPSFRDLALRVDQIRDNMAG
mutant_tumor   LLKNNGRLPRPDGCPDEIYMIMTECWNNNVNQRPSFRDLALRVDQIRDNMAG
               ***************************************************
            1081        .         .         .         .         . 1132

Call:
  read.fasta(file = "A18541014_mutant_seq.fa")

Class:
  fasta

Alignment dimensions:
  2 sequence rows; 1132 position columns (1132 non-gap, 0 gap)

+ attr: id, ali, call
```

Q2. What are the tumor specific mutations in this particular case (e.g. A130V)?

We could score residue conservation then find the non 1.0 scoring position. These will be the mutation positions:

```r
conserv(seq)
```

```
  [1]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [15]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [29]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [43]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [57]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [71]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [85]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [99]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[113]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[127]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[141]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[155]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[169]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[183]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[197]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[211]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[225]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[239]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[253]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[267]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[281]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[295]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[309]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[323]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[337]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[351]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[365]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[379]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[393]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[407]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[421]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[435]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[449]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[463]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[477]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[491]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[505]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[519]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[533]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[547]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
```

```
 [561]  1.0  1.0  1.0  1.0 -0.3  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [575]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [589]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [603]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [617]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [631]  1.0  1.0 -0.1  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [645]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [659]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [673]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [687]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [701]  1.0  1.0  1.0  1.0  1.0  1.0  1.0 -0.1  1.0  1.0  1.0  1.0  1.0  1.0
 [715]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [729]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [743]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [757]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [771]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [785]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [799]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [813]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [827]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [841]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [855]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [869]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [883]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [897]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [911]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [925]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [939]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [953]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [967]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [981]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
 [995]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1009]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1023]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1037]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1051]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1065]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1079]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1093]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1107]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
[1121]  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
```

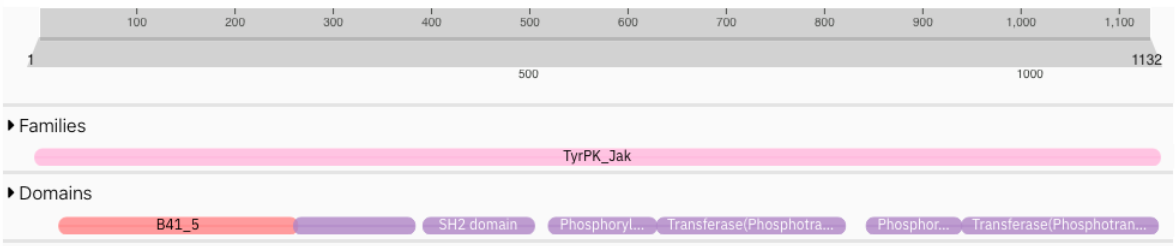
```r
mutation.sites <- which(conserv(seq) < 1)
```

```r
paste(seq$ali[1, mutation.sites],
mutation.sites,
seq$ali[2, mutation.sites])
```

```
[1] "R 565 V" "S 633 E" "P 708 R"
```

> Q3. Do your mutations cluster to any particular domain and if so give the name and PFAM id of this domain? Alternately note whether your protein is single domain and provide it's PFAM id/accession and name (e.g. PF00613 and PI3Ka).

Domains from InterPro search: (https://www.ebi.ac.uk/interpro/result/InterProScan/iprscan5-R20260310-182219-0230-34038458-p1m/internal-1773166935369-54-1/)

PFAM: 

Description & Coordinates: "R 565 V" - Phosphorylase Kinase; domain 1 (521-628) and Transferase(Phosphotransferase) domain 1 (629-816)

"S 633 E" - Transferase(Phosphotransferase) domain 1 (629-816)

"P 708 R" - Transferase(Phosphotransferase) domain 1 (629-816)

Use this website for Q4-5 (https://portal.gdc.cancer.gov/genes/ENSG00000096968)

> Q4. Using the NCI-GDC list the observed top 2 missense mutations in this protein (amino acid substitutions)?

JAK2 R683G - chr9:g.5078360A>G JAK2 T875N - chr9:g.5089726C>A

> Q5. What two TCGA projects have the most cases affected by mutations of this gene? (Give the TCGA "code" and "Project Name" for example "TCGA-BRCA" and "Breast Invasive Carcinoma").

TCGA-UCEC - Uterine Corpus Endometrial Carcinoma TARGET-AML - Acute Myeloid Leukemia

Q6. List one RCSB PDB identifier with 100% identity to the wt_healthy sequence and detail the percent coverage of your query sequence for this known structure? Alternately, provide the most similar in sequence PDB structure along with it's percent identity, coverage and E-value. Does this structure "cover" (i.e. include or span the amino acid residue positions) of your previously identified tumor specific mutations?

Chain A, Tyrosine-protein kinase JAK2 [Homo sapiens] - Query Coverage: **42%** - Percent Identity: **100.00%** - E-value: **0.0**